

SEG800 Signal Cheatsheet

Module 1 Week 1: slides01_Intro.Signals (1).pdf — Module 2 Week 2: slides02.Systems.pdf

1 Signals and Systems Basics

Signal: a function of one or more independent variables (e.g., time, space).
System: hardware/software device (or algorithm) that operates on an input signal to produce an output signal.
Algorithm: method/rules implementing the system's mathematical operation(s).

2 Signal Classifications (Module 1 Week 1)

2.1 Continuous-Time vs. Discrete-Time

Continuous-time (analog): defined for every real time value.
Discrete-time: defined only at specific instants (often indexed by integer n); undefined otherwise.
Sampling: selecting values of an analog signal at discrete times.

2.2 Continuous-Valued vs. Discrete-Valued

Continuous-valued: amplitude can be any real number.
Discrete-valued: amplitude limited to discrete levels (as stored/represented in a computer).
Digital signal: discrete-time + discrete-valued.
Quantization: continuous-valued $\rightarrow$ discrete-valued (e.g., rounding/truncation).

2.3 Deterministic vs. Random

Deterministic: uniquely described by an explicit expression, table, or well-defined rule.
Random: unpredictable/uncertain; not practically described by a simple explicit rule.

2.4 Discrete-Time Signal Representations

Functional form $x[n]$, tabular values, sequence notation, graphical (stem plot).

2.5 Elementary Discrete-Time Signals

Unit sample (impulse):

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Unit step:

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

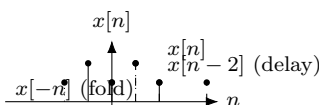
Unit ramp: $r[n] = n u[n]$.

Exponential: $x[n] = a^n$ for all n ; causal version often appears as $a^n u[n]$.
Complex exponential: $x[n] = r^n e^{j\theta n}$.

2.6 Common Time/Amplitude Operations

Time shift: $x[n - k]$ (delay if $k > 0$, advance if $k < 0$).
Folding (reflection): $x[-n]$.
Important: addition and time shifting are **not** commutative.
Down-sampling / time scaling: $x[\mu n]$ (integer μ).
Amplitude ops: Addition: $y[n] = x_1[n] + x_2[n]$. Multiplication: $y[n] = x_1[n]x_2[n]$. Scaling: $y[n] = A x[n]$.

Tiny stem-plot example (time shift + fold):



2.7 Energy vs. Power (Discrete-Time)

Energy:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Average power:

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Energy signal: $0 < E < \infty$ (then $P = 0$).

Power signal: $0 < P < \infty$.

2.8 Periodic vs. Aperiodic

Periodic: $\exists N \in \mathbb{Z}, N > 0$ such that $x[n + N] = x[n]$.

Smallest such N is the **fundamental period**.

Discrete-time sinusoid is periodic iff its normalized frequency is rational (as stated in the slides).

Periodic energy/power: nonzero periodic signals are **power** signals (energy = ∞).

Average power for period N :

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

2.9 Even/Odd Signals and Decomposition

Even: $x[n] = x[-n]$. **Odd:** $x[n] = -x[-n]$.

$$x[n] = x_e[n] + x_o[n]$$

$$x_e[n] = \frac{x[n] + x[-n]}{2}, \quad x_o[n] = \frac{x[n] - x[-n]}{2}$$

3 Discrete-Time Systems (Module 2 Week 2)

Discrete-time system: maps input $x[n]$ to output $y[n]$ via a rule:

$$y[n] = T\{x[n]\}$$

3.1 Basic System Blocks

Adder: $y[n] = x_1[n] + x_2[n]$.

Constant multiplier: $y[n] = a x[n]$.

Signal multiplier: $y[n] = x_1[n]x_2[n]$.

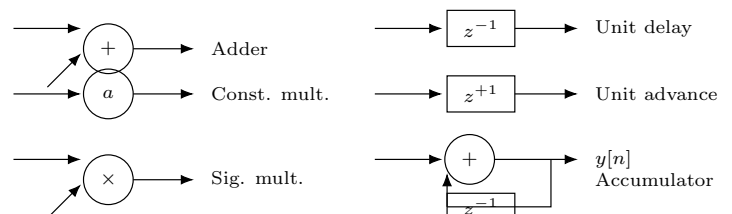
Unit delay: $y[n] = x[n - 1]$.

Unit advance: $y[n] = x[n + 1]$.

Accumulator: $y[n] = y[n - 1] + x[n]$.

Initially relaxed: system has zero initial condition / stored state, e.g., $y[n_0] = 0$.

Block diagram symbols (common):



3.2 System Classification Checklist

Memoryless (static): $y[n]$ depends only on $x[n]$. Often written as $y[n] = T\{x[n], n\}$.

Dynamic: depends on past/future samples (has memory). Finite memory (duration N): depends only on $\{x[n], x[n-1], \dots, x[n-N]\}$. Infinite memory: otherwise. Depending on future samples requires a stored signal (not physically realizable in real-time).

Time-invariant (initially relaxed): for *every* input $x[n]$ and *every* integer shift k ,

$$T\{x[n-k]\} = y[n-k]$$

Linear (superposition): for constants a, b ,

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$

Scaling: $T\{ax[n]\} = aT\{x[n]\}$. Additive: $T\{x_1[n] + x_2[n]\} = T\{x_1[n]\} + T\{x_2[n]\}$.

Causal: depends only on present/past inputs. Often written $y[n] = F(x[n], x[n-1], x[n-2], \dots)$.

BIBO stable: if $|x[n]| \leq M_x < \infty$ for all n , then $|y[n]| \leq M_y < \infty$ for all n .

3.3 Interconnection of Discrete-Time Systems

Cascade (series): $y_1[n] = T_1\{x[n]\}$, $y[n] = T_2\{y_1[n]\} = T_2\{T_1\{x[n]\}\} = (T_2 \circ T_1)\{x[n]\}$.

Parallel: $y[n] = y_1[n] + y_2[n] = T_1\{x[n]\} + T_2\{x[n]\}$.

4 LTI Systems and Convolution (Module 2 Week 2)

4.1 Impulse Resolution of a Sequence

Any discrete-time signal can be expressed as a weighted sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

4.2 Impulse Response and Convolution Sum

For LTI systems, define impulse response:

$$h[n] = T\{\delta[n]\}$$

Then the output is:

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

Where convolution comes from: impulse decomposition + superposition (linearity) + time invariance.

4.3 Manual Convolution Procedure

Folding (time-reverse), shifting, multiplication, summation (choose the easier sequence to fold).

Commutative: $x * h = h * x$.

4.4 Convolution Properties (LTI)

Identity: $x[n] * \delta[n] = x[n]$.

Shift: $x[n] * \delta[n-n_0] = x[n-n_0]$.

Commutative: $x * h = h * x$.

Associative: $(x * h_1) * h_2 = x * (h_1 * h_2)$.

Distributive: $x * (h_1 + h_2) = x * h_1 + x * h_2$.

4.5 Causality and Stability via $h[n]$

LTI causal $\Leftrightarrow h[n] = 0$ for $n < 0$.

Initially relaxed LTI BIBO stable $\Leftrightarrow h[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

5 FIR vs. IIR and Difference Equations (Module 2 Week 2)

FIR: finite-duration impulse response (finite memory).

IIR: infinite-duration impulse response (often due to feedback; infinite memory).

Difference equations are commonly used to represent LTI systems (especially IIR).

5.1 General Difference Equation (LTI)

$$y[n] = - \sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

FIR: $a_k = 0$ (no feedback). IIR: some $a_k \neq 0$ (feedback).

6 Smoothing Filters Mentioned (Module 2 Week 2)

Causal moving average (MA), length M :

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k]$$

Symmetric moving average (SMA): non-causal (uses future samples).

First-order exponential smoothing (typical form):

$$y[n] = a y[n-1] + (1-a) x[n]$$

(slides note $b = 1 - a$).